LOCATING-DOMINATING SETS IN HEXAGONAL STRIPS: MINIMUM DENSITY 1/3 EXACTLY AT WIDTHS DIVISIBLE BY 3

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Abstract. The minimum density of a locating-dominating set in the infinite hexagonal strip of width k equals $1/3$ if and only if 3 divides k. Gomes and Wakabayashi proved equality for widths divisible by three. We prove the converse and show that the excess above $1/3$ is $\Theta(k^{-2})$ as k grows in either of the other two residue classes. The lower bound follows by removing three rows at a time from a boundary with no local defect.

For every width we give an explicit periodic locating-dominating set whose density equals the value proposed in Conjecture 9 of Gomes and Wakabayashi. At widths 7, 8 and 10, we prove matching lower bounds for arbitrary sets satisfying the locating-domination conditions, including nonperiodic sets. The resulting minimum densities are 26/77, 27/80 and 47/140, respectively. The exact formula for general widths outside the multiples of three remains open.

1. The width question and main results

A locating-dominating set S places sensors at vertices of a graph. Each vertex outside S must have a nonempty set of neighboring sensors, and these sets must distinguish every two vertices outside S. On the infinite hexagonal grid, Honkala and Laihonen proved that the minimum sensor density is $1/3$. When does restricting the grid to a strip increase this minimum? [1]

Strip coordinates. We use the row–column convention of the circular bars of Gomes and Wakabayashi.[1] Write (i, j) for row $i \in \{1, \dots, k\}$ and column $j \in \mathbb{Z}$. Consecutive columns in each row are adjacent. The edge from (i, j) to $(i + 1, j)$ is present exactly when $i < k$ and $i + j$ is odd. The alternating vertical edges have this phase because $(1, 1)$ has degree two. Our coordinates are

$$\Phi(i, j) = (x, y) = (j, i - 1), \qquad \Phi^{-1}(x, y) = (y + 1, x).$$

Horizontal neighbors map to $(x \pm 1, y)$. Since $x + y = i + j - 1$, the other neighbor maps to $(x, y + (-1)^{x+y})$ whenever its row lies in $0 \le y < k$. This bijection preserves adjacency in both directions, sends each row to a row, and sends a translation by two columns to $x \mapsto x + 2$. We use the resulting strip H_k throughout. [2, §3 and Figure 6]

Density. For $a \in \mathbb{Z}$ and $L \ge 1$ let $I(a, L) = \{a, \dots, a + L - 1\} \times \{0, \dots, k - 1\}$. For a locating-dominating set S define its uniform lower column density by

$$\underline{d}_{\mathrm{col}}(S) = \liminf_{L\to\infty} \inf_{a\in\mathbb{Z}} \frac{|S \cap I(a, L)|}{kL}.$$

Our lower bounds apply to this quantity. To compare with the density in the conjecture, let $B_R(w)$ be the closed graph ball about $w = (x_0, y_0)$ and put

$$d_k = \inf_S \inf_{w\in V(H_k)} \limsup_{R\to\infty} \frac{|S \cap B_R(w)|}{|B_R(w)|}.$$

Here S ranges over all locating-dominating sets. Omitting the center gives the open-ball convention of Gomes and Wakabayashi and has no effect on the limit superior. [2, §1]

For fixed k, set $J_R = I(x_0 - R, 2R + 1)$ and $h = 3(k - 1)$. For $R \ge h$ we have $J_{R-h} \subseteq B_R(w) \subseteq J_R$. Indeed, to reach a target row one can cross at most $k - 1$ interfaces, taking at

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[1] In the full-grid edge formula in §1 of their paper, the first coordinate runs along a row. In their bar construction, the first coordinate indexes the row. We follow the latter convention throughout, including in the comparison with Theorem 8 and Conjecture 9. [2, §§1–4]

most one horizontal step to correct parity before each crossing. This takes at most $2(k-1)$ steps and changes the column by at most $k-1$. Moving along the target row then gives a path to (x, y) of length at most $|x - x_0| + h$. The other inclusion follows because each step changes the column by at most one. Thus $|J_R \setminus B_R(w)| \le 2kh$, independently of S and w. For fixed k, the inclusions give $|B_R(w)| = k(2R+1)+O_k(1)$ and hence $|B_{R+1}(w)|/|B_R(w)| \to 1$. This is the slow growth property studied for identifying codes on hexagonal strips by Sampaio, Sobral and Wakabayashi [3, Definition 3.1]. The same inclusions compare the locating-dominating density in balls and column intervals. More explicitly,

$$\frac{|S \cap B_R(w)|}{|B_R(w)|} \ge \frac{|S \cap J_{R-h}|}{|J_R|} = \frac{|S \cap J_{R-h}|}{|J_{R-h}|} \, \frac{2(R-h)+1}{2R+1}.$$

For fixed k, the last factor tends to one. The first factor is at least the infimum over starting columns for its length, so

$$\liminf_{R\to\infty} \frac{|S \cap B_R(w)|}{|B_R(w)|} \ge \underline{d}_{\mathrm{col}}(S)$$

for every center w. The same lower bound holds for the ball limsup. Any bound on $\underline{d}_{\mathrm{col}}(S)$ valid for every locating-dominating set S therefore survives both infima in the definition of d_k. Periodic constructions supply the upper bounds.

Gomes and Wakabayashi proved that $d_k = 1/3$ at every width divisible by three. At other widths they proposed an exact formula for d_k. We determine which widths have the plane density and bound the cost of the strip boundary at every remaining width. [2, Theorem 8 and Conjecture 9]

Theorem 1 (Width classification). For every integer $k \ge 1$, $d_k = 1/3$ if and only if $3 \mid k$. If $k = 3n + r$ with $n \ge 0$ and $r \in \{1, 2\}$, put $c_1 = 4$ and $c_2 = 2$. Then

$$\frac{1}{6k(12n+6)} \;\le\; d_k - \frac{1}{3} \;\le\; \frac{1}{3k(k+c_r)}.$$

Here $12n + 6$ is $4k + 2$ for $r = 1$ and $4k - 2$ for $r = 2$. Thus $d_k - 1/3 = \Theta(k^{-2})$ as $k \to \infty$ within either nonzero residue class modulo three. In each class the upper bound comes from a periodic set with the density proposed in Conjecture 9.

Theorem 2 (Three exact widths). The minimum densities at widths 7, 8 and 10 are

$$d_7 = \frac{26}{77}, \qquad d_8 = \frac{27}{80}, \qquad d_{10} = \frac{47}{140}.$$

Gomes and Wakabayashi had already reported periodic sets with all three densities. We prove the corresponding lower bounds, which apply also to nonperiodic sets. Together with their Theorem 8 this determines d_k at every width $1 \le k \le 10$. Combining these results, the widths still to be settled in Conjecture 9 are those with $k \ge 11$ and $3 \nmid k$. [2, Theorem 8 and the paragraph following Conjecture 9]

2. A nonnegative defect at each site

Fix a locating-dominating set S in H_k. Write $t(v) = |N(v) \cap S|$, with $N(v)$ the open neighborhood in the full strip. An unselected vertex u with $t(u) = 1$ is a *private neighbor* of its unique sensor. For $v \in S$, let $p(v)$ count its private neighbors. Two private neighbors of the same sensor would have identical signatures, so $p(v) \le 1$. Conversely, suppose every unselected vertex is dominated. If two distinct unselected vertices have the same singleton signature, both are private neighbors of its unique sensor. If their common signature contains two or more sensors, two of those sensors and the two vertices form a four-cycle. The strip has no four-cycle, so under domination the condition $p(v) \le 1$ at every sensor suffices for location.

Define the *defect* at v by

$$E(v) = \begin{cases} (3 - \deg(v)) + t(v) + (1 - p(v)), & v \in S, \\ \max\{t(v) - 2, 0\}, & v \notin S. \end{cases}$$

Every summand is a nonnegative integer. A site is *perfect* if its defect is zero. A perfect sensor has degree three, no neighboring sensor and exactly one private neighbor. A perfect unselected site sees one or two sensors. On the bottom row at width $k \geq 2$, the even columns have degree three and the odd columns have degree two. Only the latter are excluded as perfect sensors by degree alone. At width one every vertex has degree two.

Lemma (Defect identity). Let $I = I(a, L)$ consist of L consecutive columns and put $m = |S \cap I|$. Then

$$\left| \sum_{v \in I} E(v) - (6m - 2kL) \right| \leq 4k.$$

On a full horizontal period that preserves the strip and the sensor pattern, the error is zero.

Proof. Orient each edge between a sensor and an unselected vertex u toward u. Give it weight two if $t(u) = 1$ and weight one otherwise. A sensor v sends total weight $\deg(v) - t(v) + p(v)$, so its defect is four minus its outgoing weight. An unselected vertex u receives total weight $t(u) + \mathbf{1}_{\{t(u)=1\}}$. Subtracting two gives its defect in each of the three cases $t(u) = 1, 2, 3$.

Summing over I cancels every internal transfer. With F_{in} and F_{out} the total weights entering and leaving I, respectively, the remaining balance is

$$\sum_{v \in I} E(v) = 4m - 2(kL - m) + F_{\text{in}} - F_{\text{out}}.$$

Exactly $2k$ horizontal edges cross the two ends of I, and each carries at most two units. This bounds the absolute error by $4k$, uniformly in a, L and S. Over a full period the two end cuts are translates and their signed contributions cancel. All degrees and private-neighbor counts in this argument are evaluated in the full strip. □

Fix k and a positive integer ℓ. Suppose a locating-dominating set S satisfies $\sum_{v \in I(a,\ell)} E(v) \geq 1$ for every starting column a. Partition the first $\ell \lfloor L/\ell \rfloor$ columns of $I(a, L)$ into disjoint intervals of length ℓ. The remaining columns have nonnegative defect. Hence

$$6|S \cap I(a, L)| - 2kL \geq \sum_{v \in I(a,L)} E(v) - 4k \geq \lfloor L/\ell \rfloor - 4k.$$

Divide by $6kL$, take the infimum over a, and then take the lower limit as $L \to \infty$. This gives

$$\underline{d}_{\text{col}}(S) \geq \frac{1}{3} + \frac{1}{6k\ell}.$$

For a bound on d_k, the same $\ell = \ell(k)$ must work for every locating-dominating set S and every a. We shall prove this for $k = 3n + 1$ and $k = 3n + 2$ with $\ell = 12n + 6$.

3. Three rows peel off at a perfect boundary

Assume $k \geq 3$ and $E(v) = 0$ at every vertex of H_k. We first describe the bottom three rows of the entire strip. The finite-window version needed for the density bound is proved in the next section.

Three-row peeling lemma. Up to horizontal translation by an even number of columns, the sensors in rows $0, 1, 2$ have one of the two patterns below. The entries are selected column residues modulo six. Figure 1 shows the same two patterns. If row 3 is present, every edge between rows 2 and 3 has two unselected endpoints. Removing the bottom triple leaves all sites of the remaining strip perfect.

Pattern	Row 0	Row 1	Row 2
Dense boundary	$0, 2, 4$	none	$1, 3, 5$
Sparse boundary	$2, 4$	$0, 3$	$1, 5$

Proof. On row 0 only even sites can be sensors. Let b_i be the indicator of a sensor at $(2i, 0)$. The intervening odd site must be dominated, so consecutive zeros are impossible. The bit at

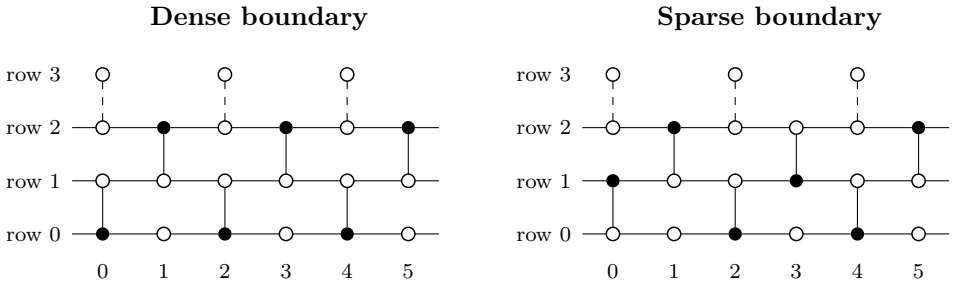


FIGURE 1. One six-column period of each perfect boundary. Filled circles are sensors and open circles are unselected sites. When row 3 exists, only its cut-edge endpoints are shown. Each dashed edge joins two open circles, so removing the bottom three rows preserves the sensors seen from above.

$(2i, 1)$ is $1 - b_i$. If the bottom site is selected, independence forbids the site above it. Otherwise that site is its only possible sensor.

Suppose $b_i = 0$. The sensor at $(2i, 1)$ already has a private neighbor below, and both its horizontal neighbors are unselected. We have $b_{i+1} = 1$, and $(2i + 1, 0)$ is private to this bottom sensor. The unselected site $(2i + 2, 1)$ must therefore see a second sensor, necessarily $(2i + 3, 1)$. Independence now gives a zero at $(2i + 4, 1)$, hence $b_{i+2} = 1$. The site above this last bottom sensor already sees two sensors, as does its left neighbor on row 0. Its right neighbor must be private, forcing $b_{i+3} = 0$. Reflection about column $2i$ gives the same recurrence to the left. Iterating in both directions on the bi-infinite sequence shows that either every b_i is one or its zeros occupy exactly one residue class of indices modulo three. This conclusion uses no periodicity assumption. Fix a zero at column 0 in the latter case.

In the dense case both horizontal neighbors of each bottom sensor see two sensors. Its neighbor above must be private. This leaves all of row 1 unselected and forces the odd sites of row 2 to be selected to dominate the odd sites of row 1. Independence excludes the even sites of row 2. Each such even site sees both adjacent sensors, so an additional sensor above it would violate perfection.

In the sparse case the recurrence fixes row 1 as in the table. Its sites at residues 1 and 5 see the row-1 sensor at residue 0, which already has its private neighbor below. Their second sensors must lie directly above them at residues 1 and 5 of row 2. Independence excludes their horizontal neighbors and the site above the row-1 sensor at residue 3. This gives row 2 as displayed. Its vertex at residue 0 sees two sensors. The row-2 sensors at residues 1 and 5 must have private neighbors at residues 2 and 4, respectively, since their other neighbors each see two sensors. At residue 0, a sensor in row 3 would give the row-2 vertex three sensor neighbors. At residues 2 and 4, it would destroy the only private neighbor of the adjacent row-2 sensor. This covers every even residue. In both patterns the edges between rows 2 and 3, when present, therefore have two unselected endpoints.

Deleting edges with two unselected endpoints changes no vertex's set of neighboring sensors. It also preserves each sensor's degree and private neighbors. The remaining sites therefore retain defect zero. To write the shorter strip in our original parity convention, use $(x, y) \mapsto (x+1, y-3)$ on $y \geq 3$. □

Conversely, each pattern in the table is perfect on H_3, as one checks on its six columns. The dense pattern is unchanged by an even translation, while the sparse pattern has three even translates. In the correct parity frame these triples may be stacked independently, since all newly introduced edges join unselected endpoints. For $k = 3n$, these are exactly the 4^n configurations in fixed coordinates for which $E(v) = 0$ at every vertex. Each has horizontal period dividing six. By the defect identity, they have density $1/3$ and exhaust the periodic locating-dominating sets of that density. The class of all locating-dominating sets of density $1/3$ is larger. For example, adding a sensor at a degree-two boundary site of a perfect stack preserves its density.

It also preserves location, since signatures of the remaining unselected vertices still differ on the original sensors. The new sensor has positive defect.

4. Quantitative peeling and uniform density

Finite-window lemma. Let $k = 3n + r$ with $n \geq 0$ and $r \in \{1, 2\}$. In every locating-dominating set on H_k, every interval of $12n + 6$ consecutive columns contains a site of positive defect.

Proof. We shall cut a strip repeatedly, so we use the following local tests on its current vertex labels. A selected site must have degree three, no selected neighbor, and exactly one private neighbor. An unselected site must have one or two selected neighbors. Call a site passing its test *perfect*. On the original locating-dominating set this is equivalent to $E(v) = 0$ by Section 2. The tests still make sense on the remaining rows after a cut, regardless of the labels outside the interval.

A cut in nine columns. Suppose the current height is $h \geq 4$ and x is even in the current brick coordinates. If all sites in rows $0, 1, 2$ and columns $x - 4, \ldots, x + 4$ are perfect, then both $(x, 2)$ and $(x, 3)$ are unselected. These are the endpoints of a vertical edge, and even x exhausts the edges across this row boundary. Translation by an even number of columns reduces the check to $x = 0$.

Here is the finite calculation used in this implication. Introduce a selection bit for each site read by a local test. For a selected center v, a neighbor w is private precisely when w and every neighbor of w other than v are unselected. These tests read at most two graph steps from a center. We impose them at the 27 centers only, leave the surrounding labels free subject to those tests, and require at least one of $(0, 2), (0, 3)$ to be selected. The resulting formula is unsatisfiable. The two relevant heights are 4 and 5. At height at least 5, all sites and adjacencies read by the tests are the same as at height 5, since no test reads above row 4. The supplied checker constructs the clauses from the coordinates and refutes them by unit propagation and branching on both values of a bit. The following counts include the terminal checks explained below. `check_local.py`

Height	Tested rows	Columns	Additional constraint	Clauses	Search nodes
4	0–2	$[-4, 4]$	at least one cut endpoint selected	530	19
5	0–2	$[-4, 4]$	at least one cut endpoint selected	530	19
1	0	$[-1, 1]$	none	36	1
2	0–1	$[-2, 2]$	none	170	3

One peeling step. Suppose all sites in columns $[a, b]$ are perfect, with $b - a + 1 \geq 18$. Delete the bottom three rows and restrict attention to $[a + 6, b - 6]$. For every cut edge in $[a + 4, b - 4]$, its nine-column window lies in $[a, b]$, so both endpoints are unselected. A test at a retained site in $[a + 6, b - 6]$ sees only columns in $[a + 4, b - 4]$. Removing these edges changes the degree only of unselected endpoints and leaves every relevant set of neighboring sensors unchanged. Every test centered in $[a + 6, b - 6]$ therefore passes after the cut. Only the labels and adjacencies within two steps of those centers enter the next peeling step. Reindex the surviving rows by $(x, y) \mapsto (x + 1, y - 3)$ to restore the brick convention. The interval shifts by one and loses twelve columns in length.

The last six columns. At height one, three consecutive perfect sites are impossible. All three have degree two and must be unselected, leaving the middle site undominated. At height two, consider the five columns $[-2, 2]$. Vertical edges join the two rows at even columns only. If all sites here were perfect, all odd-column sites would be unselected. If both endpoints at column 0 were unselected, neither would see a sensor, since their horizontal neighbors lie in odd columns. Independence excludes selecting both endpoints. Thus exactly one endpoint is selected. The opposite endpoint would be private to that sensor. On the sensor's row, the site in column 1 would then need a second sensor in column 2. This would force the other row's site in

column 2 to be unselected, leaving that row's site in column 1 undominated. This contradiction proves the height-two terminal claim. Any six consecutive columns contain five consecutive columns with even center, and hence also contain the requisite three-column window.

Now suppose all $12n + 6$ initial columns had defect zero. After j cuts the current height is $3(n - j) + r$ and the perfect interval has length $12(n - j) + 6$. For $j < n$ the height is at least four and the length at least eighteen, so another cut is valid. At $j = n$ we have six perfect columns at height r, contrary to the terminal claim. This also handles $n = 0$ directly. Hence every initial interval contains positive defect. □

The window length is independent of the locating-dominating set and the starting column. Applying the interval bound of Section 2 and the graph-ball comparison of Section 1 gives

$$d_k \geq \frac{1}{3} + \frac{1}{6k(12n + 6)} \qquad (k = 3n + r,\ r = 1, 2).$$

5. Periodic constructions at the conjectured densities

For $k = 3n + r$, $r \in \{1, 2\}$, set $c_1 = 4$, $c_2 = 2$ and $L = 2(k + c_r)$. We construct a periodic locating-dominating set of density

$$\frac{1}{3} + \frac{1}{3k(k + c_r)}.$$

A construction at this density proves the upper bound in Theorem 1. Its optimality at widths $7, 8, 10$ requires the lower bounds in the next section. For $3 \mid k$ the perfect stacks of Section 3 already attain $1/3$.

Row words. A one marks a sensor. Number the rows from zero and group them as $3j + a$ with $a = 0, 1, 2$, stopping at row $k - 1$. In group j, rotate each length-L word to the right by $3j$ columns and repeat it periodically. The following table gives the unrotated words in increasing order of a within each group. Each row denotes the concatenation PA^uMB^vT, with an en dash for an empty string. For $r = 1$ and $n \geq 2$, the exponents (u, v) are $(n - 2, 0)$ in group 0, $(n - 1 - j, j - 1)$ in an interior group, and $(0, n - 2)$ in group n. For $r = 2$ and $n \geq 1$, they are $(n - 1, 0)$ in group 0 and $(n - j, j - 1)$ thereafter. In the last group of this second family use only the first two words.

r	Group	P	A	M	B	T
1	first	10100100	101000	–	–	10100010100100
1	first	00010010	010010	–	–	01001001001001
1	first	01001001	000101	–	–	00010100010010
1	interior	00100100	101010	1010010010	010101	0101
1	interior	01001001	000000	0001001001	010101	0101
1	interior	00010010	010101	0100100100	000000	0000
1	last	001001010010010101	–	–	010101	0101
2	first	10100100101	000101	–	–	000
2	first	01001010010	010010	–	–	010
2	first	00001010000	101000	–	–	101
2	later	00100100101	000101	–	001001	001
2	later	10001010010	010010	–	010100	010
2	later	10001010000	101000	–	010100	010

The exceptions are $k = 1, 2, 4$. The table gives one periodic word for each row of the strip. Blank cells indicate rows outside the strip.

Width	Row 0	Row 1	Row 2	Row 3
1	0010100101			
2	10010010	10100100		
4	1010010100101010	0001010101000000	0100000000010101	0010101010100101

The supplementary generator implements these row rules for the two parametric families. `construction.py`

Counting sensors. Count ones before rotation. For $r = 1$ the first three rows contain L ones, each of the $n - 1$ interior groups contains $L - 1$, and the last row contains $3n + 3$. For $r = 2$ each of the n full groups contains L and the last two rows together contain $4n + 6$. Thus the number m of sensors in one period is

$$r = 1: \quad m = L + (n-1)(L-1) + (3n+3) = \frac{(3n+1)L+2}{3},$$
$$r = 2: \quad m = nL + (4n+6) = \frac{(3n+2)L+2}{3}.$$

The three small patterns also satisfy $3m = kL + 2$ by direct count. Since $L = 2(k + c_r)$, this gives $m/(kL) = 1/3 + 1/[3k(k + c_r)]$.

Legality at every width. A failure would have a local witness. Domination requires a sensor in each closed neighborhood. On this triangle-free graph, adjacent unselected vertices with nonempty signatures cannot have the same signature, and vertices at distance greater than two have disjoint neighborhoods. For two centers at distance two it suffices to hit the symmetric difference of their closed neighborhoods. If either center is selected, that center itself meets the requirement. Up to reversal, the displacements to test are $(2, 0)$, $(-1, 1)$ and $(1, 1)$, with both centers in the strip.

We now show why a finite check of the row words suffices. Suppose a requirement fails at a center $(x, 3j + a)$, and write $u_0 = x - 3j \pmod{L}$ for its unrotated column. Every queried in-strip site has displacement (i, b) with $|i| \le 3$, $|b| \le 2$. Its group is $g = j + \lfloor (a + b)/3 \rfloor$, so $|g - j| \le 1$, and its column in that group's word is $u_0 + i - 3(g - j) \pmod{L}$. Thus all queried positions lie in the same cyclic arc of at most thirteen positions, $u_0 - 6, \ldots, u_0 + 6$.

For $n > 22$ at least one of the following shortenings applies. Positions in a word are numbered from zero.

- If $n - j \ge 12$, decrease n by one and keep j fixed. The relevant groups retain their roles. Each queried word loses one copy of its first repeated motif. The corresponding runs have a common start, position 8 for $r = 1$ or 11 for $r = 2$, and contain at least ten copies each. The terminal group cannot be queried in this case.
- If $j \ge 12$, decrease both n and j by one. Each queried group g is now $g - 1$ in the shorter strip. The first motif count is unchanged, and each word loses one copy from its last repeated run. For $r = 1$ this run begins at $18 + 6(n - 1 - g)$ in an interior group and at 18 in the terminal group. For $r = 2$ it begins at $11 + 6(n - g)$. In either family the starts among the queried words differ by at most twelve positions and the runs have length at least sixty.

In either case the repeated runs share a linear interval of at least 48 positions. The protected cyclic arc occupies at most thirteen positions and meets this interval in at most two pieces, even when it crosses the period seam. Its removal leaves at least 35 positions in at most three pieces. One piece therefore contains at least six consecutive positions. Delete the same six-position block D there in every queried word. Within a six-periodic run, deletion of any six consecutive bits gives the same word as shortening the run by one repeat.

To track the witness, put $L' = L - 6$ and let π_D renumber the surviving positions from zero in the shorter word. It fixes positions before D and subtracts six from those after D. For $|t| \le 6$ we have $\pi_D((u_0 + t) \bmod L) = (\pi_D(u_0) + t) \bmod L'$, since the intervening arc is disjoint from D. Write $j' = j$ for the first shortening and $j' = j - 1$ for the second. A center in row $3j + a$

and unrotated column u_0 maps to

$$(x', y') = (3j' + \pi_D(u_0),\, 3j' + a)$$

in the shorter periodic strip. In particular, the second shortening shifts the group frame by $(-3, -3)$ before the column renumbering. Both the frame shift and the removal of six columns preserve brick parity.

Every queried group shifts by the same group offset as j, so the formula for the protected arc preserves the bit at each local displacement. In the first case the lower boundary stays fixed and the upper boundary is beyond the queried groups, even after shortening. In the second case the distance to the upper boundary stays fixed and the lower boundary is beyond the queried groups. Thus the same local requirement would fail in the shorter strip.

Iterating would give a failure at $n \leq 22$. The fixed check uses the original grid coordinates at each such width. A pair with equal nonempty signatures must have centers at distance at most two. Translate one center into columns $0, \ldots, L-1$ by a multiple of L. The other then lies in columns $-2, \ldots, L+1$. The checker reads sensor bits modulo L but compares signatures as sets of absolute vertex coordinates, covering comparisons across the period seam. The check finds all signatures nonempty and distinct, both in the finite range and for the three exceptional patterns. `check_construction.py`

The proof above uses shortening, the finite direct checks and the small patterns. As a supplementary computational check, a separate transcription expresses the same word rules as integer constraints with n unbounded. Splitting by r, the row offset a, and $u_0 \bmod 6$ gives $2 \cdot 3 \cdot 6 = 36$ instances. Each asks whether a local requirement fails. Both Z3 and cvc5 returned unsatisfiable on every instance. `check_symbolic.py`

This proves the construction bound for all widths. Combining it with Section 4 gives $d_k - 1/3 = \Theta(k^{-2})$ as k tends to infinity in either nonzero residue class modulo three.

6. Exact densities at widths 7, 8 and 10

Section 5 supplies a periodic set at each proposed density. To obtain the opposite inequality for every locating-dominating set, we use a separate integer certificate for each width. The certificate assigns a potential to the states encountered when the strip is read two columns at a time.

States and transitions. Order the vertices (x, y) by $kx + y$. At the first site of each domination or separation requirement from Section 5, insert the corresponding positive clause. A one satisfies all pending clauses containing the current site. A zero removes the current site from those clauses, and an empty clause rejects the path. Express the surviving clauses by their offsets from the next unread site. If one clause contains another, satisfying the smaller clause suffices for both. The state is the sorted family of inclusion-minimal pending clauses. The empty family means that no obligation is pending. The checker regenerates the clauses from the original grid neighborhoods, at both column parities. Two columns restore the initial phase.

Fix an even starting column and start with the empty family. Omitting obligations introduced strictly to the left of the cut can only enlarge the set of finite paths. In any genuine locating-dominating configuration, each clause introduced from the cut onwards contains a selected site. Its scan therefore never rejects, and keeping only inclusion-minimal pending clauses preserves the path. This gives a legal finite path for every scan length and every genuine configuration.

Finite verification. For the chosen k, the certificate gives integers h, c and a potential P on a finite set A of states at aligned cuts. After a header specifying k, h, c and the state count, the certificate stores one integer potential value per state. To attach these values to states, the checker reconstructs A from the empty family by two-column steps, verifies the count and the presence of the empty family, and sorts the states lexicographically. It reads the values in that order, regenerates the legal two-column steps from A, and checks that their set of endpoints is exactly A. For each endpoint t it computes the minimum of $P(s) + hb$ over paths from all

$s \in A$ to t, where b is the number of selected sites in the two columns. It keeps the minimum at each intermediate state after each bit, since future transitions depend only on that state and the scan phase. It checks that the resulting minimum at every t is at least $P(t) + c$.
Consequently each such path satisfies

$$hb \geq c + P(t) - P(s).$$

The verifier uses integer arithmetic and reconstructs the transitions without calling the optimization program. The last column of the table records the range of the certified potentials. `check_fixed.cpp`

k	Cut states	h	c	$c/(2kh)$	$[\min P, \max P]$
7	18,538	11	52	26/77	$[-4, 66]$
8	72,667	5	27	27/80	$[0, 36]$
10	1,140,699	14	94	47/140	$[-12, 112]$

Return to the infinite strip. On N consecutive aligned steps containing a total of m sensors, summation yields

$$hm \geq Nc + P(s_N) - P(s_0) \geq Nc - (\max P - \min P).$$

These steps cover $2kN$ sites. In an arbitrary interval of L columns, discard at most one column at either end to obtain an even starting column and an even length $2N$. The discarded columns contain a nonnegative number of sensors. Put $\Delta P = \max P - \min P$. Using $2N \geq L - 2$, we obtain for every origin a and every $L \geq 2$

$$\frac{|S \cap I(a, L)|}{kL} \geq \frac{c}{2kh} - \frac{c + \Delta P}{hkL}.$$

The error is independent of both a and S. Taking the infimum over a and then the lower limit in L gives $\underline{d}_{\mathrm{col}}(S) \geq c/(2kh)$. Section 1 transfers the bound to graph balls for every locating-dominating set, including nonperiodic ones. The periodic sets in Section 5 give equality in all three cases and complete the proof of Theorem 2.

7. What remains of Conjecture 9

The strip has the plane's minimum locating-dominating density exactly at the widths divisible by three. At the other widths we know the order of the excess and have a periodic construction at the conjectured value. Together with the previously known exact values, Theorem 2 settles every width up to ten. The remaining question is whether the construction is optimal at every $k \geq 11$ with $3 \nmid k$.

Fix one of the remaining widths $k = 3n + r$, where $r \in \{1, 2\}$. Put $W = k + c_r$, with $c_1 = 4$ and $c_2 = 2$. For a periodic locating-dominating set, let L be an even horizontal period, m the number of sensors in one period, and $e = 3m - kL$. Doubling an odd period doubles both m and L and leaves the density unchanged. The density comparison for this particular set is exactly

$$\frac{m}{kL} - \frac{1}{3} = \frac{e}{3kL}, \qquad \frac{m}{kL} \geq \frac{1}{3} + \frac{1}{3kW} \quad \Longleftrightarrow \quad We \geq L.$$

We claim that Conjecture 9 at width k holds if and only if $We \geq L$ for every periodic locating-dominating set on H_k. If the conjecture holds, every periodic set has density at least the conjectured minimum, so the displayed comparison gives $We \geq L$. In particular, a single periodic set with $We < L$ would give a construction below the proposed value. Conversely, suppose the inequality holds for every periodic set. A periodic set attains the unrestricted minimum at this fixed width, so applying the inequality to that set gives $d_k \geq 1/3 + 1/(3kW)$. The construction in Section 5 gives the matching upper bound, proving equality. [2, §2, Theorem 3(ii)]

A local sufficient condition would require at least two defect units in every window of W consecutive columns. To see this, let E_j be the sum of the defects in column j, indexed modulo L. Section 2 gives $\sum_{j=0}^{L-1} E_j = 6m - 2kL = 2e$. Summing the L window totals then gives

$$\sum_{a=0}^{L-1} \sum_{i=0}^{W-1} E_{(a+i) \bmod L} = W \sum_{j=0}^{L-1} E_j = 2We.$$

For each offset i, the L starts visit every column exactly once. This also counts repeated columns when $W > L$. If each window contributed at least two, the left side would be at least $2L$, implying $We \geq L$.

The settled width $k = 10$ illustrates the distinction. Here $W = 14$, and the supplementary witness has zero defect on columns $0, \ldots, 13$. It specifies all ten rows on columns $-4, \ldots, 17$ and selects every site outside this patch. Appending four fully selected columns and repeating the resulting 26 columns gives a periodic locating-dominating set. Nonselected vertices in different copies are at horizontal distance at least five, and the protected window retains its two-step neighborhood. This set has $m = 153$ sensors per period, so $e = 3 \cdot 153 - 10 \cdot 26 = 199$ and the total defect is 398. All of it lies outside the displayed window. Thus the same set has a window with zero defect and satisfies the period inequality with $We = 14 \cdot 199 = 2786 > 26 = L$. `check_counterwindow.py`

The example rules out the two-unit window requirement as a uniform statement across strip widths. For $k \geq 11$ with $3 \nmid k$, the open question is whether the total defect of every period satisfies $We \geq L$.

Supplementary files. The named checkers, row-word generator, and fixed certificates are in `HONEYCOMB_STRIP_CONJECTURE9_CERTIFICATES_20261103_v2.zip`. See `REPLAY.md` in the archive for the verification commands.

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